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Deep learning classification of systemic sclerosis from multi-site photoplethysmography signals

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Abstract

Introduction: A pilot study assessing a novel approach to identify patients with Systemic Sclerosis (SSc) using deep learning analysis of multi-site photoplethysmography (PPG) waveforms ("DL-PPG").

Methods: PPG recordings having baseline, unilateral arm pressure cuff

occlusion and reactive hyperaemia flush phases from 6 body sites were studied in 51 Controls and 20 SSc patients. RGB scalogram images were obtained from the PPG, using the continuous wavelet transform (CWT). 2 different pre-trained convolutional neural networks (CNNs, namely, GoogLeNet and EfficientNetB0) were trained to classify the SSc and Control groups, evaluating their performance using 10-fold stratified cross validation (CV). Their classification performance (i.e., accuracy, sensitivity, and specificity, with 95% confidence intervals) was also compared to traditional machine learning (ML), i.e., Linear Discriminant Analysis (LDA) and K-Nearest Neighbour (KNN).

Results: On a participant basis DL-PPG accuracy, sensitivity and specificity for GoogLeNet were 83.1 (72.3–90.9), 75.0 (50.9–91.3) and 86.3 (73.7–94.3)% respectively, and for EfficientNetB0 were 87.3 (77.2–94.0), 80.0 (56.3–94.3) and 90.1 (78.6–96.7)%. The corresponding results for ML classification using LDA were 66.2 (53.9–77.0), 65.0 (40.8–84.6) and 66.7 (52.1–79.2)% respectively, and for KNN were 76.1 (64.5–85.4), 40.0 (19.1–63.9), and 90.2 (78.6–96.7)% respectively.

Discussion: This study shows the potential of DL-PPG classification using CNNs to detect SSc. EfficientNetB0 gave an overall improved performance compared to GoogLeNet, with both CNNs performing better than the traditional ML methods tested. Our automatic AI approach, using transfer learning, could offer significant benefits for SSc diagnostics in a variety of clinical settings where low-cost portable and easy-to-use diagnostics can be beneficial

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Keywords

- deep learning
- machine learning
- photoplethysmography
- pulse
- Raynaud’s
- scleroderma
- systemic sclerosis

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